

Dylan Pieper

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Profile

Data scientist with 5+ years engineering end-to-end data systems, primarily for the health sciences. Engineers pipelines, databases, statistical models, reports, dashboards, and advanced analytic applications. Skilled at translating messy, large, and complex data into meaningful decisions for psychology, education, health, and government audiences.

Skills

- **Methods:** Bayesian/frequentist multilevel modeling, ML/LLM/NLP applications, psychometrics, mixed methods
- **Languages:** R, Python, SQL (MySQL, PostgreSQL, DuckDB), HTML/CSS/JS
- **Applications:** Shiny, Quarto, Tableau, Power BI, REDCap, Qualtrics
- **Infrastructure:** Git, GitHub, Docker, Azure, Microsoft Foundry, Posit Connect, Linux (RHEL)

Experience

Research Analyst

Nov 2025 - Present

CISR-UME, School of Medicine and Public Health, University of Wisconsin - Madison, WI

- **Led** data strategy for medical education research projects, partnering with faculty, administrative staff, and IT professionals to translate research questions and project protocols into data extraction and analytic workflows across curriculum innovation, assessment, and training systems
- **Engineered** an end-to-end machine learning pipeline on GitLab, Azure, and Microsoft Foundry, integrating multiple data sources to transcribe OSCE video recordings (LearningSpace API, Whisper), apply NLP to clean and analyze transcripts, and generate quality-checked pedagogical feedback at scale
- **Engineered** a second end-to-end machine learning pipeline for first-pass grading of open-ended assessment responses, with reproducible workflows, version control, and documentation on GitLab, Azure, and Microsoft Foundry
- **Designed and deployed** a faculty review web interface in Quarto and HTML/JS for evaluating AI-generated feedback, with a secure JSON-based offline data collection workflow that meets regulatory requirements for sensitive OSCE data
- **Designed and deployed** the Medical Education Research Hub, a production dashboard on Posit Connect that aggregates project metadata via the Google Sheets API and serves as the Medical Education Office's central research directory (<https://connect.doit.wisc.edu/meded-research-hub/>)
- **Extracted, integrated, and visualized** learner content interaction data through the Learning Pool API for ad hoc reporting
- **Automated** reproducible data reporting with GitLab and Quarto, delivering sample and cohort summaries, statistics, and visualizations to faculty and administrative staff
- **Designed and deployed** the unit website (<https://meded.labs.wisc.edu/>)

Experience

Data Scientist

Sep 2021 - Oct 2025

School of Pharmacy, University of Pittsburgh - Pittsburgh, PA

- **Led** data strategy across the lifecycle of grant-funded health research projects, partnering with implementation specialists, researchers, university and clinical IT departments, care providers, and public officials to translate project protocols and partner needs into data extraction and analytic workflows
 - **Designed** health record and REDCap integrations for a state-funded opioid treatment initiative
 - **Engineered** ETL pipelines in R and DuckDB to consolidate complex clinical, operational, and research data across a 250+ site treatment network
 - **Automated** reproducible data reporting with GitLab and Quarto for data quality validation and treatment-standard setting, delivering patient population summaries, statistics, and visualizations to clinicians and directors
 - **Applied** Bayesian multilevel modeling to identify predictors of patient retention in care management
 - **Wrangled** complex administrative court data across three counties for an Arnold Ventures criminal justice project
 - **Applied** frequentist multilevel models to evaluate racial disparities in pretrial bail, and created interactive D3 Sankey visualizations tracking case outcomes from arraignment through sentencing
 - **Designed** a MySQL relational database for centralized project management and implementation tracking, developing optimized and maintainable SQL queries that linked operational and geographic data across studies
 - **Designed and deployed** data collection tools (REDCap, Qualtrics) and business intelligence dashboards (Shiny, Tableau) for project teams, treatment sites, and government partners, including a Tableau dashboard for the DEA to monitor suspicious drug prescribing patterns
 - **Mentored** junior data staff across recruitment, onboarding, and professional development
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Field Interviewer

Aug 2021 - Nov 2021

NORC, University of Chicago - Chippewa Falls, Wisconsin

- **Conducted** structured interviews across diverse locations, maintaining data quality through standardized protocols while upholding confidentiality and research ethics standards
 - **Managed** end-to-end logistics including scheduling, travel, and participant communication to ensure successful data collection across assigned areas
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Research Specialist

Aug 2020 - Aug 2021

Michele Gelfand's Culture Lab, University of Maryland - College Park, MD

- **Engineered and published** frequentist multilevel models of COVID-19 outcomes across 57 countries in The Lancet Planetary Health
 - **Designed and deployed** a Shiny dashboard to detect and compare threatening language across texts (<https://www.michelegelfand.com/threat-dictionary>)
 - **Conducted** randomized experiments on message framing for wearing masks using Qualtrics
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Writing Coach

Aug 2017 - May 2020

University of Northern Iowa - Cedar Falls, IA

- **Coached** students in writing and statistics and supported ELLs and students with learning disabilities
- **Trained** new tutors and developed training materials

Education

Master of Arts - Social Psychology

2018 - 2020

University of Northern Iowa - Cedar Falls, Iowa

- Completed courses Statistical Computing I and II in R
 - Worked as a Research Assistant in the Psychoneuroendocrinology Lab
 - Worked as a Teaching Assistant in undergraduate psychology courses (Introduction to Psychology, Biopsychology, and Psychology of Gender)
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Bachelor of Arts - Psychology

2014 - 2018

University of Northern Iowa - Cedar Falls, Iowa

Publications

- Gelfand, M. J., Jackson, J. C., Pan, X., Nau, D., **Pieper, D.**, Denison, E., Dagher, M., Van Lange, P. A. M., Chiu C., & Wang, M. (2021). The relationship between cultural tightness-looseness and COVID-19 cases and deaths: A global analysis. *The Lancet: Planetary Health*. [https://doi.org/10.1016/S2542-5196\(20\)30301-6](https://doi.org/10.1016/S2542-5196(20)30301-6)
- Gelfand, M. J., Li, R., Stamkou, E., Denison, E., Fernandez, J., Choi, V., Chatmen, J., Jackson, J. C., **Pieper, D.**, & Dimant, E. (2021). Persuading conservatives and liberals to comply with mask-wearing: An intervention tournament. *Journal of Experimental Social Psychology*. <https://doi.org/10.1016%2Fj.jesp.2022.104299>
- Dimant, E., Clemente, E. G., **Pieper, D.**, & Azevedo, F. (2022). Politicizing mask-wearing: Predicting the success of behavioral interventions among Republicans and Democrats in the U.S. *Scientific Reports*, 12, 7575. <https://doi.org/10.1038/s41598-022-10524-1>
- **Pieper, D.** (2020). Challenging social systems under the threat of pollution: Replication and extension of Eadeh and Chang (2019) [Master's Thesis, University of Northern Iowa]. *UNI ScholarWorks*. <https://scholarworks.uni.edu/etd/1028>

Presentations

- **Pieper, D.** (2025, September). From messy to meaningful data: LLM-powered classification in R. *posit::conf*. <https://github.com/dylanpieper/posit-25>
- **Pieper, D.** (2025, June). redquack: An R package for memory efficient REDCap-to-DuckDB workflows. *R/Medicine*. <https://r-consortium.org/posts/redquack-an-r-package-for-memory-efficient-redcap-to-duckdb-workflows>
- Katie, W. & **Pieper, D.** (2024, June). The Pennsylvania medications for opioid use disorder (PA MOUD) technical assistance and quality improvement expansion project. *Rural Health Information Hub*. (Online resource). <https://www.ruralhealthinfo.org/project-examples/1139>
- **Pieper, D.**, Gelfand, M. J., & Denison, E., (2021, February). COVID-19: Meta-norms to wear masks and social distance in the U.S. *Society for Personality and Social Psychology*. (Remote poster session). <https://osf.io/y8cug>
- **Pieper, D.**, Hirano, H., & Harton, H. C. (2020, April). The effect of environmental threats and terrorism/violent-crime threats on political orientation and attitudes. *Midwestern Psychological Association Conference*. Chicago, Illinois. (Conference canceled) ... Also (2020) at the *University of Northern Iowa's College of Social and Behavioral Sciences Conference*. Cedar Falls, Iowa. (Conference canceled). <https://osf.io/3n7r8>
- **Pieper, D.** (2020, February). Challenging the system under the threat of pollution: Low system justification mediates a liberal political shift. *Society for Personality and Social Psychology*. New Orleans, LA. (Poster session) ... Also (2020) at the *University of Northern Iowa's College of Social and Behavioral Sciences Conference*. Cedar Falls, Iowa. (Conference canceled). <https://osf.io/v7nez>

Presentations

- **Pieper, D.** (2019, April). What's in a name? Measuring the potential influence of medical cannabis labels on attitudes toward its use. *Iowa Psychological Association Conference, Des Moines, Iowa.* (Poster session) ... Also (2019, April) at the *University of Northern Iowa's College of Social and Behavioral Sciences (CSBS) Conference.* Cedar Falls, Iowa. Awarded Best Oral Presentation (2nd place, \$250). <https://scholarworks.uni.edu/agss/2019/all/18>
- **Pieper, D.,** DeSoto, C., & Hitlan, R. (2018, April). The relations between personality and source of dietary protein. *Midwestern Psychological Association Conference.* Chicago, Illinois. (Poster session) ... Also (2017, April) at the *University of Northern Iowa's College of Social and Behavioral Sciences Conference.* Cedar Falls, Iowa.

R Packages

- **redquack:** Transfer REDCap data to a database and use in R without exceeding available memory (<https://dylanpieper.github.io/redquack>)